MAG = MAGNETOMETERS
MT = MAGNETORQUERS

Main OreSat Card Connector

J1
J-SAMTEC-TFM-120-01-L-D-RA

1 MAG_SDA
3 MAG_SCL
4 MAG_PWR
12 OPD_SDA
13 OPD_SCL
15 OPD_PWR
17 CAN2_H
20 CAN2_L
24 PWR_FLAG
25 VBUS
C1 10u 0805
28 PWR_FLAG GND
37 C3-UART-TX
38 C3-UART-RX

OreSat Power Domain signals
(See power sheet)

DOUBLECHECK

Magnetometers + IMU

MAG_SDA
MAG_SCL
IMU_INT2
IMU_INT1
MAG_EN
MAG_FAULT
MAG_SCL_uC
MAG_SDA_uC
MAG_READY

File: oresat-mag-card-sch-sensors.kicad_sch

Card Power Domain

OPD_SDA
OPD_SCL
OPD_PWR
C3-UART-TX
C3-UART-RX
CARD-UART-TX
CARD-UART-RX
BOOT_CONFIG

File: oresat-mag-card-sch-power.kicad_sch

Microcontroller

BOOT_CONFIG
MAG_READY
MAG_SDA_uC
MAG_SCL_uC
MAG_FAULT
MAG_EN
IMU_INT1
IMU_INT2
CAN2_H
CAN2_L
SWRESET
SWDIO
SWCLK
SWO
CARD_UART_TX
CARD_UART_RX
MT_X_ISENSE
MT_Y_ISENSE
MT_Z_ISENSE
MT_Z_PHASE
MT_Z_PWM
MT_Y_PHASE
MT_Y_PWM
MT_X_PHASE
MT_X_PWM
MT_ILIM
MT_EN/FAULT
MT_STBY/RST
MT_EN

File: oresat-mag-card-sch-microcontroller.kicad_sch

Debug Connector

J2
SM14B-SHLS-TF
VBUSP
3.3V
DEVICE-D+
DEVICE-D-
VTARGET
NRST
SWDIO
SWCLK
SWO
CARD-TX
CARD-RX
DEBUG-0
TP1
PAD
GND

Magnetorquers

MT_X_ISENSE
MT_Y_ISENSE
MT_Z_ISENSE
MT_Z_PHASE
MT_Z_PWM
MT_Y_PHASE
MT_Y_PWM
MT_X_PHASE
MT_X_PWM
MT_ILIM
MT_EN/FAULT
MT_STBY/RST
MT_EN
MT_X_OUT_BD
MT_X_OUT_AD
MT_Y_OUT_BD
MT_Y_OUT_AD
MT_Z_OUT_BD
MT_Z_OUT_AD

X: MT_X_OUT_B 1, MT_X_OUT_A 2 — J5 S02B-PASK-2
Y: MT_Y_OUT_B 1, MT_Y_OUT_A 2 — J4 S02B-PASK-2
Z: MT_Z_OUT_B 1, MT_Z_OUT_A 2 — J3 S02B-PASK-2

File: oresat-mag-card-sch-magnetorquers.kicad_sch

MAGNETORQUER CONNECTORS
Use JST PAP-02V-S housing
with SPHD-002T-P0.5 contacts

Fiducials on top layer only
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial

OreSat Card v1.3 with 0x RF Connectors

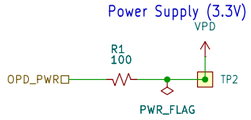
Title: ESI Mag Card
Rev: v1.1

Ethereal Space Inc.	Sheet: esi-mag-card
Date: 2026-10-19	Id: 1/5

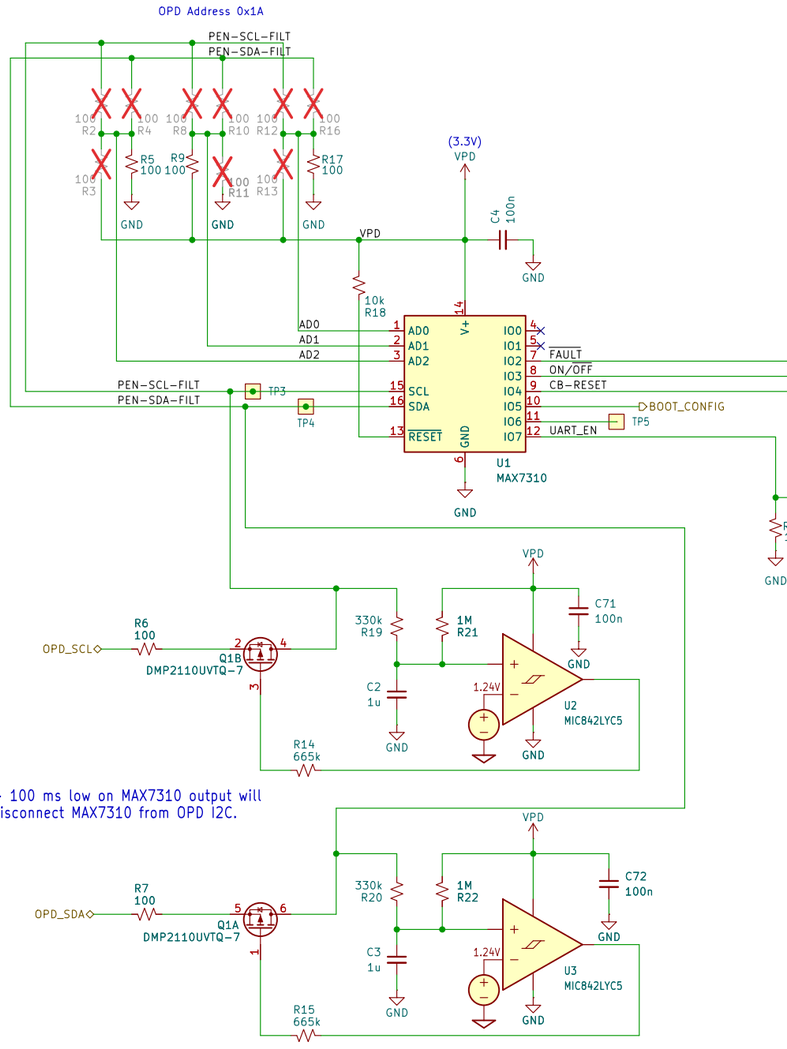
All resistors are 0603 +/- 1% and AEQ unless marked
All caps are 0603 ceramic 10% X5R >= 16V Vww and AEQ unless marked

OreSat Power Domain (OPD) Controller

OreSat Power Domain (OPD) Power Supply (3.3V)

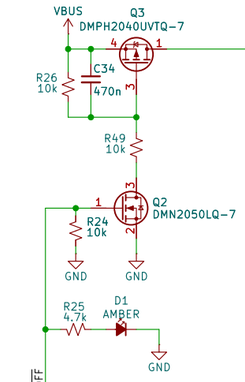


OPD Address 0x1A

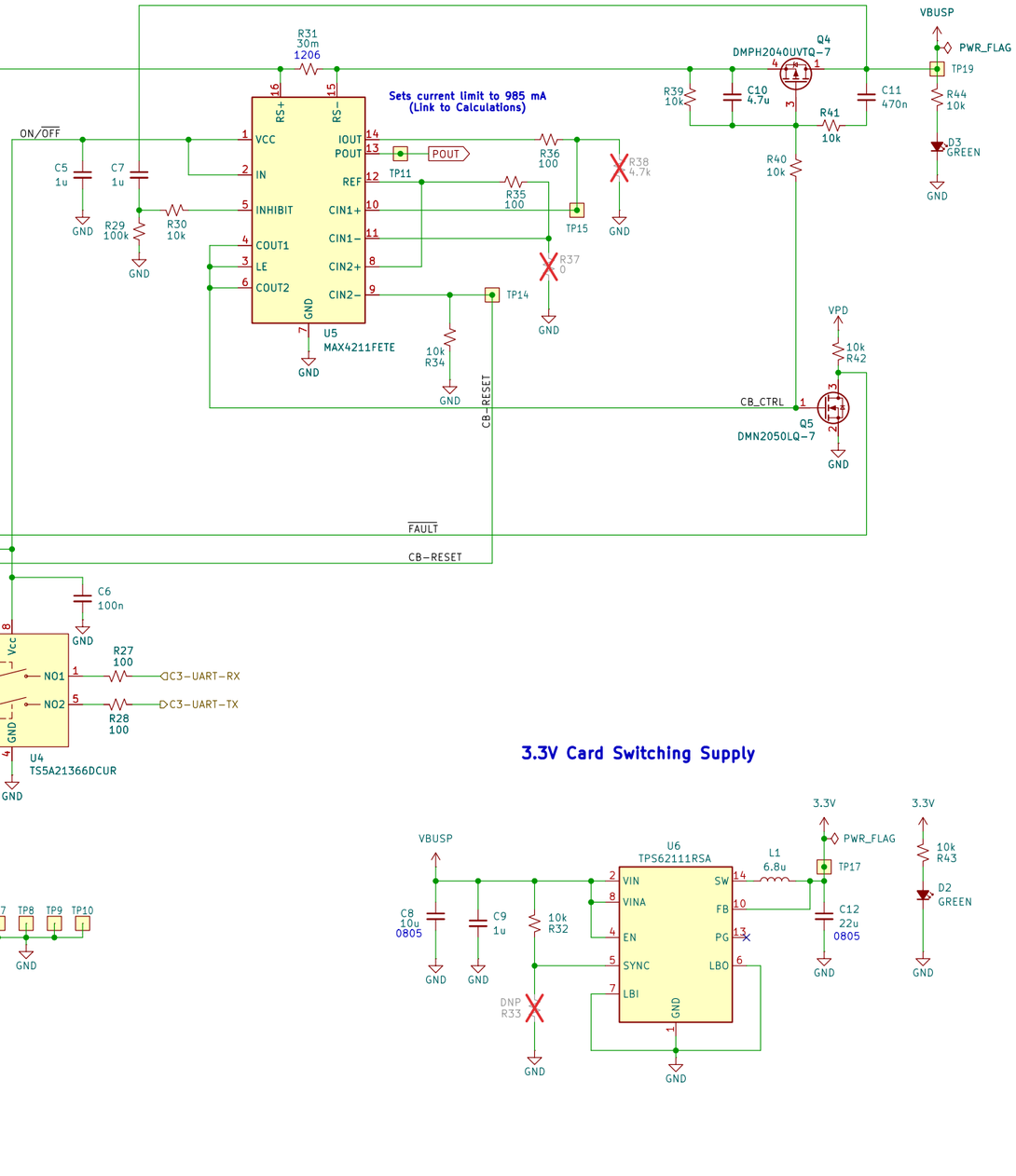


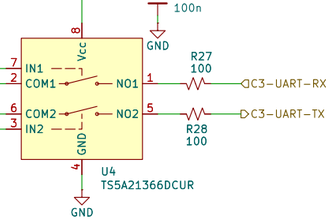
> 100 ms low on MAX7310 output will disconnect MAX7310 from OPD I2C.

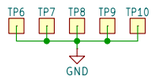
Main Power Switch
Vbus = (6.0,7.2,8.4)V



Card Power Switch and Circuit Breaker







3.3V Card Switching Supply

All resistors 0603 +/- 1% and AEQ unless marked
All caps are 0603 ceramic 10% X5R >= 16V Vww and AEQ unless marked

Title: ESI Mag Card	
Rev: v1.1	
Ethereal Space Inc. Date: 2026-10-19	Sheet: Card Power Domain Id: 2/5

ETHEREAL SPACE

U7F
MCXN947VKLT

P4_23 43
P4_21/ANA_22 42
P4_20 41
P4_19 39
P4_17/ANA_18
P4_16 38
P4_15 37
P4_13/ANA_14 36
P4_12 35
P4_7/ANA_7 29
P4_6/ANA_6 28
P4_5/ANA_5 27
ADC1_A0_ 26 ADC
25 ADC
DAC1_OUT/ADC0_B4/ADC1_B4/CMP0_IN5N/CMP1_IN5N/CMP2_IN5N
DAC0_OUT/ADC0_A4/ADC1_A4/CMP0_IN4N/CMP1_IN4N/CMP2_IN4N 24 DAC
ADC0_B0_ 23 ADC
ADC0_A0_ 22 ADC

MAG_READY
MAG_FAULT
MAG_EN
Magnetometer System Control
MT_EN
Magnetorquer Control
MT_EN/FAULT
MT_STBY/RST
MT_Z_PHASE
MT_Y_PHASE
MT_X_PHASE
MT_X_ISENSE
MT_ILIM
MT_Y_ISENSE
MT_Z_ISENSE

VBUSP
R77 10k
R78 4.7k
GND
Measure Vbusp
Vbusp = Vadc * 3.128
(8.20V = 2.622 V)

U7D
MCXN947VKLT

PWM1_B0 21
PWM1_A0 20
PWM1_B1 19
P2_4 18
P2_3 17
P2_2 16
P2_1 15
P2_0 14

MT_X_PWM
MT_Y_PWM
MT_Z_PWM
IMU_INT2
IMU_INT1

Each magnetorquer operate individually of each other
PWM specification does not really matter here.

U7E
MCXN947VKLT

P3_21 56
P3_20 57
P3_17 60
P3_16 61
P3_15 62
P3_14 63
P3_13 64
P3_12 65
P3_11 67
P3_10 68
P3_9 69
P3_8 70
P3_7 71
P3_6 72
P3_1 74
P3_0 75

3.3V
R47 10k
SWRESETD
TP20
C16 100n
GND

U7A
MCXN947VKLT

RESET_B 9
USB0_DP 46
USB0_DM 45
P0/P1
VDD 83
VDD 96
VDD_P3 59
VDD_P3 66
VDD_P3 73
VDD_P4 34
VDD_USB 44
VDD_LDO_CORE/VDD/VDD_P2 13
VDD_DCDC/VDD_LDO_SYS 54
VDD_BAT 47
VDD_ANA 30
VREFH 31
VREFL 32
VDD_SYS 55
DCDC_LX 53
VSS_DCDC 52
VSS_P4 33
EP 101
VDD_CORE 12
VDD_CORE 58

C19 4.7u
C22 1u
FB1 120 ohm
C21 2.2u
L2 1.5u
C20 22u 0805
C17 100n
C18 4.7u
PWR_FLAG
PWR_FLAG VDD_CORE

Note: Disable LDO_CORE in software for more power savings

NXP is weird, digital communication may be labeled under FlexCOMM
Look carfully for SPI UART and I2C lines. May have to dig a little.
Looking at the linked excel file will help alot.

U7B
MCXN947VKLT

TMS/SWDIO_ 76 SWDIO
TCLK/SWCLK 77 SWCLK
TDO/SWO_ 78 SWO
P0_3 79
P0_4 80
P0_5 81
ISPMODE_N 82
I2C_SDA_ 84
I2C_SCL_ 85
P0_18 86
P0_19 87
P0_20 88
P0_21 89
P0_22 90
P0_23 91

R76 10k
BOOT_CONFIG
R46 4.7k
R48 4.7k
MAG_SDA_uC
MAG_SCL_uC
I2C

Table 225 page 428 of MCXN Reference Manual.
ISPMODE_N (P0_6 pin) determines boot mode
High = Flash Memory Boot
Low = ISP Boot

Rev: 001
Hardware version number
(use pullups)

REV (P0_23:21)	Mag Card Version
111	v1.0
001	v1.1

U7G
MCXN947VKLT

P5_0 48
P5_1 49
P5_2 50
P5_3 51

U7C
MCXN947VKLT

P1_0 92
P1_1 93
P1_2 94
P1_3 95
P1_4 97
P1_5 98
P1_6 99
P1_7 100
ISP-UART_RXD 1 CARD_UART_RX
ISP-UART_TXD 2 CARD_UART_TX
CAN0_TXD 3 CAN-TXD
CAN0_RXD 4 CAN-RXD
P1_12 5 TCAN-nFAULT-OC
FLEXIO0_D21_ 6 MCU-LED
P1_14 7 TCAN-SILENT
P1_15 8
XTAL48M_ 10
EXTAL48M_ 11

Debug UART

Y1 24 MHz
C13 4.7p
C14 4.7p
R45 4.7k
D4 AMBER
GND

CAN Transceiver

3.3V
C15 100n
U8 TCAN337G
TXD 1
RXD 4
FAULT 5
S 8
VCC 3
GND 2
CANH 7 CAN2_H
CANL 6 CAN2_L

24 MHz allows LPTMRs to work

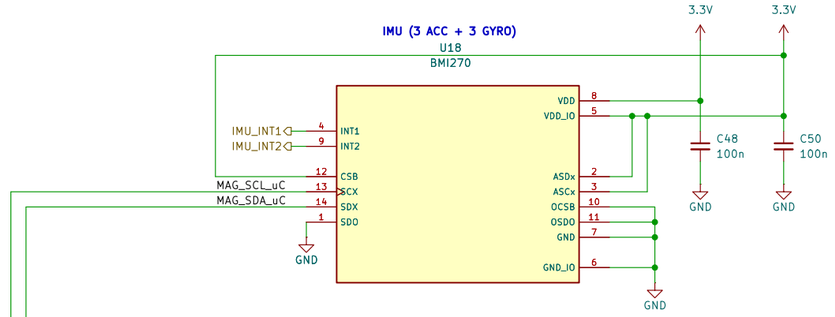
C70 C23 C24 C25 C26 C27 C28 C29 C30 100n

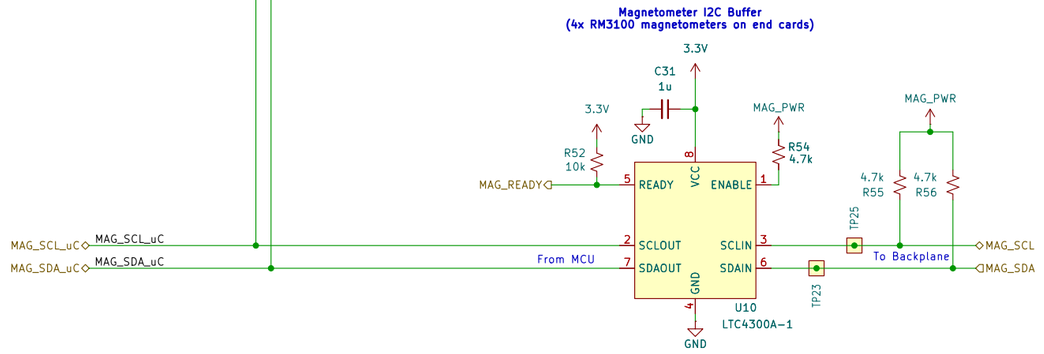
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Title: ESI Mag Card
Rev: v1.1

Ethereal Space Inc.
Date: 2026-10-19

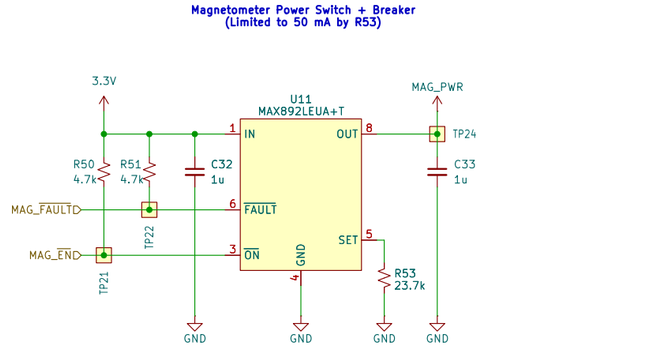
Sheet: Microcontroller
Id: 3/5





PNI RM3100 Magnetometers are located on End Caps:

"PZ1": +Z End Cap, +X−Y corner, Address 00
"PZ2": +Z End Cap, −X+Y corner, Address 01
"MZ1": −Z End Cap, +X−Y corner, Address 10
"MZ2": −Z End Cap, −X+Y corner, Address 11



Title: ESI Mag Card

Rev: v1.1

Ethereal Space Inc. Date: 2026-10-19	Sheet: Magnetometers + IMU Id: 4/5

ETHEREAL SPACE

All resistors 0603 +/- 1% and AEQ unless marked
All caps are 0603 ceramic 10% X5R >= 16V Vwv and AEQ unless marked

X-Axis Magnetorquer Driver

U12
STSPIN250

EN/FAULT
STBY/RESET
PH
PWM
REF
TOFF
TEST0
TEST1
VS
OUTA1
OUTA2
OUTB1
OUTB2
SENSEA
SENSEB
EP
GND

VBUSP
C45 22u 0805
C47 2.2u
C37 10n
C38 1n
C39 10n 16kHz
R61 1k
R63 47k
C43 22n
toff=37u
TP34 MT_X_PHASE
TP29 MT_X_PWM
TP38 MT_X_OUT_A
TP35 MT_X_OUT_B

X/Y "Rod" Magnetorquer

30 AWG with Kapton insulation
Number of wire turns: 1700 +/- 2%
Total resistance: 151 ohm +/- 10%

Imax at Vbus = 8.2 V is 54 mA
Pmax = 0.45 W

3.3V
C53 1u
R65 30m 1206
R66 30m 1206
R69 10k
TP39 MT_X_ISENSE
C57 100n
GND GND
50x gain amp

MT_EN/FAULT
From MCU
MT_EN
R60 23.7k
TP28
STSPIN pulls EN/FAULT low during failure disabling the magnetorquers
R57 10k
MT_STBY/RSTD
TP27

DAC output
MT_ILIMD
R58 23.7k
TP26
MT_REF
R59 1k

STSPIN current limits when V_{SENSE} > V_{REF}
With 3.0 V out from DAC,
V_{REF} = 0.12 V
Sense resistor gives 60 mV/A,
so max current limit is about 2 A

Z-Axis Magnetorquer Driver

U16
STSPIN250

VBUSP
C63 22u 0805
C64 2.2u
C59 10n
C60 1n
C61 10n 16kHz
R71 1k
R72 47k
C62 22n
TP43 MT_Z_PHASE
TP41 MT_Z_PWM
TP45 MT_Z_OUT_A
TP44 MT_Z_OUT_B

Z "Ring" Magnetorquer

30 AWG with Kapton insulation
Number of wire turns: 500 +/- 1%
Total resistance: 60 ohm +/- 10%

Imax at Vbus = 8.2 V is 137 mA
Pmax = 1.12 W

3.3V
C67 1u
R73 30m 1206
R74 30m 1206
R75 10k
TP46 MT_Z_ISENSE
C69 100n
GND GND
50x gain amp

Y-Axis Magnetorquer Driver

U13
STSPIN250

VBUSP
C46 22u 0805
C49 2.2u
C40 10n
C41 1n
C42 10n 16kHz
R62 1k
R64 47k
C44 22n
TP33 MT_Y_PHASE
TP30 MT_Y_PWM
TP37 MT_Y_OUT_A
TP36 MT_Y_OUT_B

X/Y "Rod" Magnetorquer

30 AWG with Kapton insulation
Number of wire turns: 1700 +/- 2%
Total resistance: 151 ohm +/- 10%

Imax at Vbus = 8.2 V is 54 mA
Pmax = 0.45 W

3.3V
C54 1u
R67 30m 1206
R68 30m 1206
R70 10k
TP40 MT_Y_ISENSE
C58 100n
U15 INA185A2Q
GND GND
50x gain amp

Title: ESI Mag Card
Rev: v1.1

Ethereal Space Inc.
Date: 2026-10-19

Sheet: Magnetorquers
Id: 5/5

ETHEREAL SPACE

All resistors 0603 +/- 1% and AEQ unless marked
All caps are 0603 ceramic 10% X5R >= 16V Vww and AEQ unless marked